# Lab: Chart Selection Decision Framework

Workshop 3 — Module 10

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Data Visualization for Data Scientists
UNYT Tirana

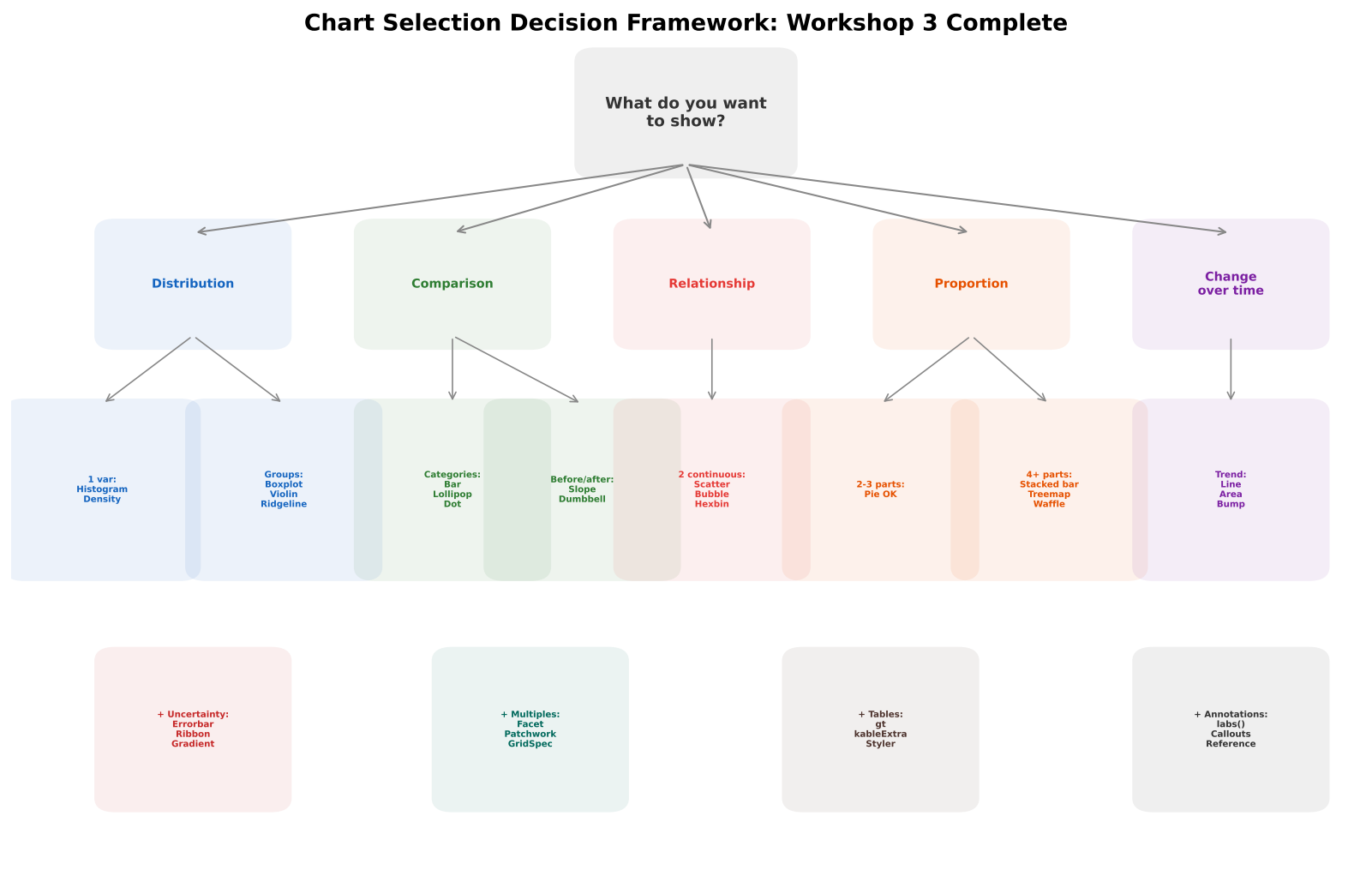
# Module Roadmap

1. The Chart Selection Decision Framework (flowchart)
2. Apply the framework to the Netflix dataset
3. Build an 8-panel dashboard using 8 chart types
4. Workshop 3 review: the complete chart encyclopedia
5. Course Project 1 introduction

**Learning Outcome**
You will apply the chart selection decision framework to a real dataset, choosing the optimal chart type for each analytical question and composing the results into a multi-panel report.

# The Decision Framework

# Chart Selection Flowchart



# The Five Questions

| Question | Chart Family | W03 |
|---|---|---|
| What is the distribution? | Histogram, density, ridgeline, box, violin | M01- |
| How do categories compare? | Bar, lollipop, Cleveland dot, dumbbell | M03 |
| What is the relationship? | Scatter, bubble, hexbin + smoothers | M04 |
| What are the proportions? | Pie, donut, treemap, waffle, stacked bar | M05 |
| How do rankings change? | Slope, bump, waterfall | M06 |
| *Enhancement layers (apply to any chart):* | | |
| How uncertain is it? | Error bars, ribbons, gradient bands | M07 |
| Multiple groups? | Facet, patchwork, GridSpec | M08 |
| Need exact values? | Styled tables, sparklines | M09 |

# Netflix Dataset

# The Netflix Dataset

| Variable | Type |
|---|---|
| type | Categorical (Movie / TV Show) |
| title | Text (8,807 unique) |
| director | Categorical (many missing) |
| country | Categorical (multi-valued) |
| date_added | Date (2008–2021) |
| release_year | Integer (1925–2021) |
| rating | Ordinal (TV-MA, TV-14, etc.) |
| duration | Numeric (minutes or seasons) |
| listed_in | Categorical (genre, multi-valued) |

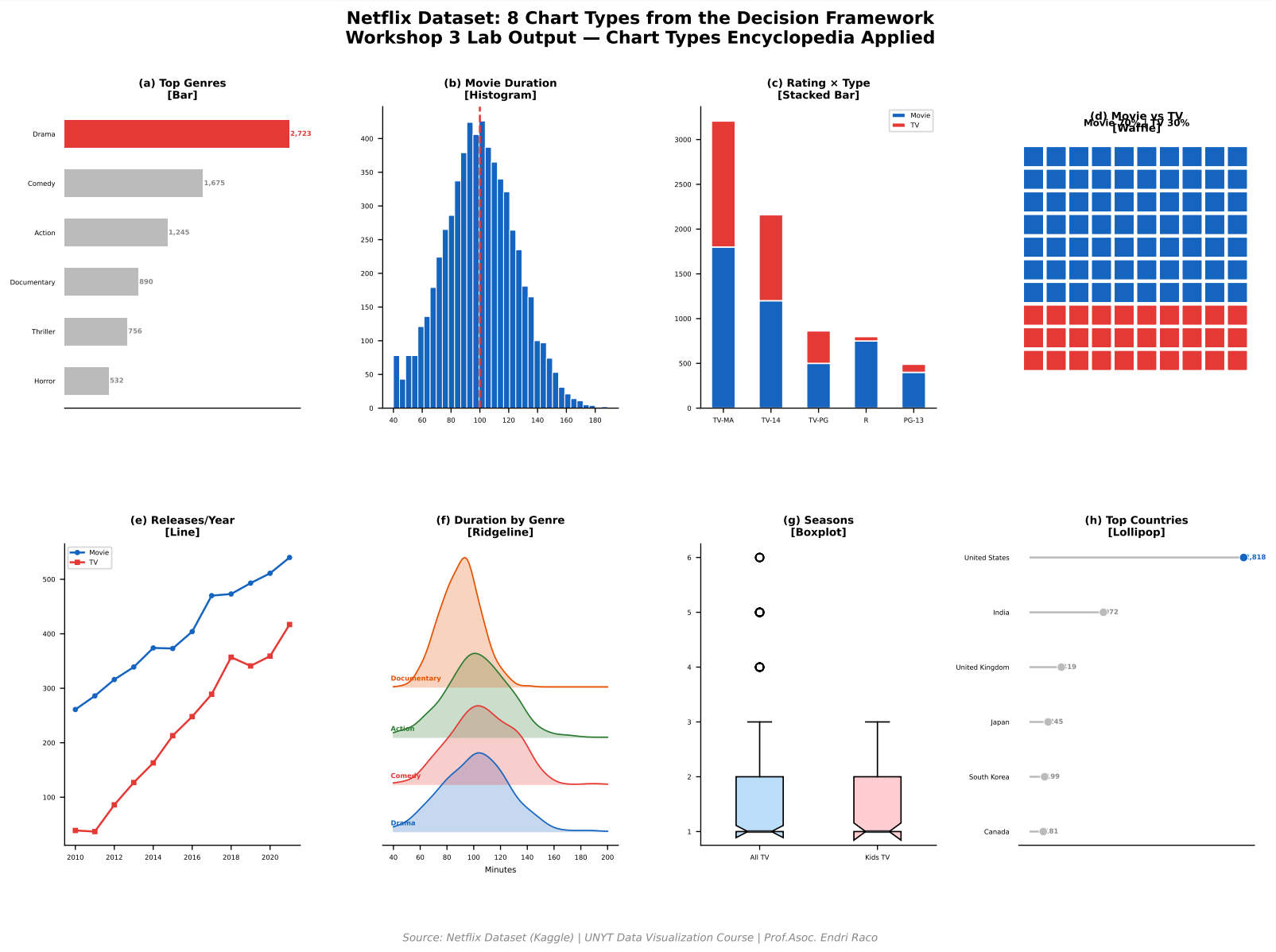
8,807 titles: 6,131 movies (70%) + 2,676 TV shows (30%). Our task: apply the decision framework to answer 8 analytical questions.

# Eight Questions → Eight Chart Types

| # | Question | Chart | Panel |
|---|---|---|---|
| 1 | Which genres dominate? | Sorted bar | (a) |
| 2 | How long are movies? | Histogram | (b) |
| 3 | Ratings by type? | Stacked bar | (c) |
| 4 | Movie vs TV split? | Waffle | (d) |
| 5 | Release trends? | Line | (e) |
| 6 | Duration by genre? | Ridgeline | (f) |
| 7 | How many seasons? | Boxplot | (g) |
| 8 | Which countries? | Lollipop | (h) |

# The Dashboard

# Netflix: 8 Chart Types in One Dashboard



# R: Building the Netflix Dashboard

```r
library(tidyverse); library(patchwork)
netflix <- read_csv("netflix.csv")

p_a <- netflix |> separate_rows(listed_in, sep=", ") |>
  count(listed_in, sort=TRUE) |> slice_max(n, n=8) |>
  mutate(listed_in = fct_reorder(listed_in, n)) |>
  ggplot(aes(x=listed_in, y=n)) + geom_col(fill="#E53935") +
  coord_flip() + theme_minimal() + labs(title="(a) Top Genres")

p_b <- netflix |> filter(type=="Movie") |>
  mutate(dur = parse_number(duration)) |>
  ggplot(aes(x=dur)) + geom_histogram(bins=35, fill="#1565C0") +
  theme_minimal() + labs(title="(b) Movie Duration")

# ... panels c through h ...

dashboard <- (p_a | p_b | p_c | p_d) / (p_e | p_f | p_g | p_h) +
  plot_annotation(
    title = "Netflix Dataset: Visual Report",
    subtitle = "Workshop 3 Lab     8 Chart Types from the Decision
        Framework",
    caption = "Source: Netflix Dataset (Kaggle) | UNYT",
    tag_levels = "a")

ggsave("netflix_report.pdf", dashboard, width=16, height=10)
```

# Python: Same Dashboard via GridSpec

```
import pandas as pd, matplotlib.pyplot as plt
from matplotlib.gridspec import GridSpec

netflix = pd.read_csv("netflix.csv")
fig = plt.figure(figsize=(16, 10))
gs = GridSpec(2, 4, figure=fig, hspace=0.45, wspace=0.35)

# (a) Sorted bar     top genres
ax_a = fig.add_subplot(gs[0, 0])
genres = netflix["listed_in"].str.split(", ").explode().value_counts
    ().head(8)
ax_a.barh(genres.index[::-1], genres.values[::-1], color="#E53935")
ax_a.set_title("(a) Top Genres")

# (b) Histogram     movie duration
ax_b = fig.add_subplot(gs[0, 1])
movies = netflix.query("type == 'Movie'")
movies["dur"] = movies["duration"].str.extract(r"(\d+)").astype(
    float)
ax_b.hist(movies["dur"].dropna(), bins=35, color="#1565C0",
    edgecolor="white")
ax_b.set_title("(b) Movie Duration")

# ... panels c through h on gs[0,2], gs[0,3], gs[1,0:4] ...

fig.suptitle("Netflix: 8 Chart Types", fontsize=13, fontweight="bold
```

# Workshop 3 Review

# Workshop 3 Complete: Chart Types Encyclopedia

## Workshop 3 Complete: Chart Types Encyclopedia

| | |
|---|---|
| M01 | **Histograms, Density, Ridgeline**<br>Bin width, KDE, bandwidth |
| M02 | **Boxplots, Violins, Beeswarm**<br>Shape blindness, raincloud |
| M03 | **Bar, Lollipop, Cleveland Dot**<br>Sorted, horizontal, ink cascade |
| M04 | **Scatter, Bubble, Hexbin**<br>Overplotting, smoothers, marginals |
| M05 | **Pie, Donut, Treemap, Waffle**<br>Angle < length, when pies work |
| M06 | **Slope, Bump, Waterfall**<br>Rankings, P&L decomposition |
| M07 | **Error Bars, CI, Gradient**<br>SE vs CI vs SD, pointrange |
| M08 | **Small Multiples, Patchwork**<br>Facet vs compose, GridSpec |
| M09 | **Tables as Visualization**<br>gt, kableExtra, sparklines |
| M10 | **Lab: Chart Selection Framework**<br>Flowchart → Netflix dashboard |

# The Complete Chart Vocabulary

| Family | Charts Covered |
|---|---|
| Distribution | Histogram, density, KDE, ridgeline, box, violin, beeswarm, raincloud |
| Comparison | Bar, lollipop, Cleveland dot, grouped, stacked, 100% stacked |
| Relationship | Scatter, bubble, hexbin, 2D density, marginal |
| Proportion | Pie, donut, treemap, waffle, 100% stacked bar |
| Ranking | Slope, bump, waterfall, dumbbell |
| Uncertainty | Errorbar, pointrange, ribbon, gradient, forest |
| Composition | Facet, patchwork, GridSpec, subplot_mosaic |
| Table | gt, kableExtra, Styler, sparklines |
| **Total** | |

# Course Project 1

# Course Project 1: Visual Redesign Portfolio

**Due: End of Workshop 3 (10% of total grade)**
Select **5 published visualizations** (news media, academic papers, corporate reports). For each:

1. Critique using Tufte's principles (W01)
2. Redesign in R (ggplot2)
3. Redesign in Python (plotnine or seaborn)
4. Write a 300-word rationale

**Deliverable**: PDF portfolio + reproducible R and Python scripts.

# Workshop 3 Homework Summary

**Due: Before Workshop 4**
Visualize the Netflix dataset using $\geq$8 distinct chart types in both R and Python:

- Apply the chart selection decision framework
- Each chart must match the analytical question
- Compose into a multi-panel dashboard
- Export as PDF + PNG (300 dpi)

Total weight: 5% of course grade.

**Next**: Workshop 4 — Exploratory Data Analysis (M01: Tukey's EDA Philosophy)